Pronouns: He/Him
Nationalities: United Kingdom, Ireland
fdonnan@ucsd.edu
fergusdonnan.github.io/
github.com/FergusDonnan
0000-0002-6460-3682

Fergus Donnan

Research Interests

Galaxy Evolution, Kinematics, AGN/Stellar Feedback, Dust Obscuration/Evolution, SMBH Growth, AGN/Star-formation, PAHs, Molecules, AGN Outflows, Accretion Disks

Education

2021–2025 **DPhil (PhD) Astrophysics**, *University of Oxford*, Oxford, UK
"The Obscured Universe: Tracing Star Formation and AGN activity across cosmic time."
Supervisors: Dimitra Rigopoulou, Ismael García-Bernete.

2016–2021 **MPhys (Hons) Astrophysics**, *University of St Andrews*, St Andrews, UK, 1st class
"Echo Mapping of AGN Accretion."
Supervisors: Keith Horne, Juan Hernandez Santisteban.

Research Positions

2025 - present **Postdoc**, *University of California San Diego*, San Diego, USA

2025 **Postdoc**, *University of Oxford*, Oxford, UK

2023 **Visiting Graduate Student**, *CAB ESAC*, Madrid, Spain

2020 **LEAPS Research Student**, *Leiden Observatory*, Leiden, Netherlands

2019 **Summer Research Project**, *University of St Andrews*, St Andrews, UK

Skills

Data Analysis **Spectroscopy**, JWST/MIRI Pipeline, IFU Data cubes (Kinematics, IFU PSF Subtraction), Spectral Fitting (SPIRIT (own software), PAHDecomp (own software), PAHFIT)

Imaging, JWST/MIRI Pipeline, aperture photometry (SourceExtractor, Photutils), PSF Subtraction, SED Fitting (BAGPIPES, CIGALE, CYGNUS)

Times Series Data, PyROA (own software), cross correlation

Interferometry, VLTI Data, ALMA data

Own Software **SPIRIT, PAHDecomp, PyROA, Running Optimal Average**

Coding Python (numpy, scipy, astropy, matplotlib, jax, numpyro etc), Fortran

Awards & Scholarships

2021 **Principal's Scholarship for Academic Excellence**, Awarded annually to the fifty final-year students, at the University of St Andrews, whose academic performances are the highest in their respective faculties.

2020 **Scott Lang Prize**, Most outstanding student in the Senior Honours class in Astronomy and Astrophysics.

2016-2021 **Deans List**, For averaging a 1st class across the academic year.

2020 **Medal (Astrophysics Fourth level)**

2019 **Medal (Astrophysics Third level)**

2019 **Carnegie Trust Vacation Scholarship**, Awarded to fund a place on an undergraduate research internship programme.

2018 **Medal (Astrophysics Second level)**

Presentations

2026 **Ensenada-San Diego Astrophysical Society meeting**, Ensenada, Mexico, Contributed Talk
Dust across cosmic scales.

2026 **University of Kansas Astro/Space Seminar**, Lawrence (Online), KS, USA, Invited Talk
Kinematics of PAH features with JWST.

2026 **Exploring the Aromatic Universe in the JWST era**, London, Canada, Contributed Talk
Spatially resolved PAH spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.

2026 **Oxford Galaxy Evolution Seminar**, Oxford, UK, Invited Talk
Organic dust grains across cosmic scales: From resolved PAH maps at cosmic noon to velocity maps of PAHs in AGN outflows

2026 **Charting Cosmic Dawn**, Copenhagen, Denmark, Contributed Talk
PAHSPECS: Spatially Resolved PAH Spectroscopy at cosmic noon with MIRI MRS

2025 **IAC Galaxy Coffee**, Tenerife, Spain, Invited Talk
Spatially resolved PAH spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.

2025 **CAB Extragalactic Meeting**, Madrid, Spain, Invited Talk
The Dust Obscured Universe: Unveiling Star-formation and AGN activity across cosmic time with mid-infrared spectroscopy.

2025 **PanDust 2025**, Tucson, AZ, USA, Contributed Poster
Spatially resolved PAH Spectroscopy in the era of JWST: From kinematics of PAHs to resolved maps at cosmic noon.

2025 **UCSD Astrophysics Colloquium**, San Diego, CA, USA, Invited Talk
The Dust Obscured Universe: Unveiling Star-formation and AGN activity across cosmic time with mid-infrared spectroscopy.

2025 **Oxford Galaxy Evolution Seminar**, Oxford, UK, Invited Talk
The Obscured Growth of SMBHs: Revealing the nature of the most obscured galaxy nuclei in the mid-infrared.

2025 **Dusting Off the Secrets of the Cosmos with PRIMA Space IR Telescope**, Marseille, France, Contributed Talk
A Census of the Most Obscured Galaxy Nuclei over Cosmic Time to be revealed by PRIMA.

2025 **Behind a curtain of dust, V**, Kiruna, Sweden, Contributed Talk
Peeling Back the Layers of Extinction of the Most Obscured Nuclei in the Era of JWST: What is their nature and evolution?

2024 **Cyprus Astrophysics Workshop 2024: Luminous Infrared galaxies in the JWST era**, Nicosia, Cyprus, Invited Talk
Peeling Back the Layers of Extinction of the Most Obscured Galaxy Nuclei in the Era of JWST.

2024 **Oxford Astrophysics Colloquium**, Oxford, UK, Early Career Showcase
The Obscured Growth of SMBHs: Peeling Back the Layers of Extinction of the Most Obscured Galaxy Nuclei in the Era of JWST.

2024 **Cosmic Odysseys**, Crete, Greece, Invited Talk
Peeling back the layers of extinction of dusty galaxies in the era of JWST.

2023 **European Astronomical Society (EAS)**, Krakow, Poland, Contributed Talk
Revealing the most obscured galaxy nuclei in the mid-infrared: JWST/MIRI Spectroscopy of VV 114.

2023 **National Astronomy Meeting (NAM)**, Cardiff, UK, Contributed Talk
The most obscured galaxy nuclei with JWST/MIRI: the hidden growth of SMBHs?

2023 **The Restless Nature of AGN: 10 Years Later**, Naples, Italy, Contributed Poster
Reverberation mapping of PG 1119+120: Testing Super-Eddington Accretion.

2023 **National Observatory of Athens, AHEAD2020**, Athens, Greece (virtual), Invited Talk
The most obscured Galactic Nuclei in the mid-iR.

2022 **RAS Meeting: JWST commissioning and first science from a UK perspective**, London UK, Contributed Talk
The Obscured Nucleus of VV 114E with JWST/MIRI Spectroscopy.

2022 **Behind a curtain of dust, IV**, Sesto, Italy, Contributed Talk
Detecting Compact Obscured Nuclei in the Mid-Infrared.

Observing Proposals

2025 **CO-I**, *Unraveling Effects of Different Modes of AGN Feedback via PAH Features* , JWST (MIRI + NIRSpec). Cycle 4, 12.2 hours, PI: L. Zhang

2024 **CO-I**, *AGN and their outflows: probing fragmentation and survival of polycyclic aromatic hydrocarbons*, JWST (NIRSpec). Cycle 3, 18.7 hours, PI: I. García-Bernete

2024 **CO-I**, *Pinpointing a VLA- invisible, bright SMG in the JWST-IRAC Deep Field*, NOEMA, Grade B, 6 hours, PI: D. Rigopoulou

2023 **CO-I**, *Unveiling the AGN-host connection with PAH molecules.*, JWST (MIRI + NIRSpec). Cycle 2, 14.3 hours, PI: I. García-Bernete

2023 **CO-I**, *A JWST high-definition view of an extremely metal-poor interstellar medium.*, JWST (MIRI). Cycle 2, 31.2 hours, PI: A. Aloisi

2022 **CO-I**, *NOEMA and JWST unveil compact obscured nuclei in the Universe*, NOEMA, Grade C, 12.2 hours, PI: I. García-Bernete

2022 **CO-I**, *NOEMA and JWST unveil compact obscured nuclei in the Universe*, NOEMA, Grade B, 26.1 hours, PI: I. García-Bernete

2022 **CO-I**, *Pinpointing a Bright SMG for JWST Monitoring of Dusty Supernovae*, NOEMA, Grade C, 12.0 hours, PI: D. Rigopoulou

2021 **CO-I**, *ALMA and JWST unveil compact obscured nuclei in the Universe*, ALMA, Grade C, 30.9 hours, PI: I. García-Bernete

Other Positions

2024-2025 **Journal Club Organiser**, *Galaxies Journal Club*, Department of Physics, University of Oxford

2022-2024 **Lab Demonstrator**, *Part B Astrophysics Practicals*, Department of Physics, University of Oxford

Publications (44, 11 First Author, h-index: 19, 902 citations)

First Author

1. **Donnan, F. R.**, Sandstrom, K., Bolatto, A., et al., (2026) submitted
   JWST Observations of Starbursts: The motion of dust - PAH kinematics of M82 and NGC 253 with JWST/MIRI spectroscopy
2. **Donnan, F. R.**, Ramos Almeida, C., González Martín, O., et al., (2026), A&A, 712, A72
   Kinematic detection of dusty outflows from active galactic nuclei: Polycyclic aromatic hydrocarbon kinematics of type 2 quasars with JWST/MIRI spectroscopy
3. **Donnan, F. R.**, Sandstrom, K., Shivaei, I., et al., (2026), ApJ, 1007, 52
   PAHSPECS: Spatially Resolved PAH Spectroscopy at cosmic noon with JWST MIRI MRS
4. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2026), MNRAS, 550, stag1181
   GATOS XIV: The first direct kinematic evidence of dusty outflows from AGN through PAH kinematics of local Seyfert galaxies with JWST
5. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., et al., (2025), JATIS, 11, 031606
   Census of the most obscured galaxy nuclei over cosmic time to be revealed by PRIMA
6. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., (2024), MNRAS, 532, 75
   The kinematics of Polycyclic Aromatic Hydrocarbons (PAHs) in Galaxies revealed by Principal Component Analysis (PCA) tomography with JWST/NIRSpec
7. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2024), MNRAS, 529, 1386
   Peeling back the layers of extinction of dusty galaxies in the era of JWST: modelling joint NIRSpec + MIRI spectra at rest-frame 1.5-28 μm
8. **Donnan, F. R.**, Hernández Santisteban, J. V., Horne, K., et al., (2023), MNRAS, 523, 545
   Testing super-eddington accretion on to a supermassive black hole: reverberation mapping of PG 1119+120
9. **Donnan, F. R.**, García-Bernete, I., Rigopoulou, D., et al., (2023), MNRAS, 519, 3691
   The obscured nucleus and shocked environment of VV 114E revealed by JWST/MIRI spectroscopy
10. **Donnan, F. R.**, Rigopoulou, D., García-Bernete, I., et al., (2023), A&A, 669, A87
    A detailed look at the most obscured galactic nuclei in the mid-infrared
11. **Donnan, F. R.**, Horne, K., Hernández Santisteban, J. V., (2021), MNRAS, 508, 5449
    Bayesian analysis of quasar light curves with a running optimal average: new time delay measurements of COSMOGRAIL gravitationally lensed quasars

Second/Third Author

12. García-Bernete, I., **Donnan, F. R.**, Rigopoulou, D., et al., (in prep)
    (GATOS N. High angular resolution observations of PAH bands in AGN with JWST NIRSpec and MIRI/MRS: An alternative PAH diagnostic diagram

13. Thatte, N., Rigopoulou, D., **Donnan, F. R.**, et al., (2025), MNRAS, 545, 2047
    The PAH 3.4 micron feature as a tracer of shielding in the Orion Bar and NGC 6240
14. García-Bernete, I., **Donnan, F. R.**, Rigopoulou, D., et al., (2025), A&A, 696, A135
    On unveiling Buried Nuclei with JWST: a technique for hunting the most obscured galaxy nuclei from local to high redshift
15. García-Bernete, I., Rigopoulou, D., **Donnan, F. R.**, et al., (2024), A&A, 691, A162
    The Galaxy Activity, Torus, and Outflow Survey (GATOS). V: Unveiling PAH survival and resilience in the circumnuclear regions of AGN with JWST
16. Rigopoulou, D., **Donnan, F. R.**, García-Bernete, I., et al., (2024), MNRAS, 532, 1598
    Polycyclic aromatic hydrocarbon emission in galaxies as seen with JWST

Co-Author

17. Zhang, L., Packham, C., Hicks, E., et al., (2026), ApJL, 1007, L52
    GATOS: Distinct Feedback Modes in AGN Central Regions Revealed by Spatially Resolved JWST Spectroscopy
18. Lofaro, C. M., Díaz-Santos, T., Shivaei, I., et al., (2026)
    PAHSPECS: Polycyclic aromatic hydrocarbon properties at cosmic noon with JWST/MIRI MRS
19. Rosario, D. J., Haidar, H., Campbell, S., et al., (2026)
    GATOS XV: A JWST/MIRI survey of extended circumnuclear dust emission in nearby Seyfert galaxies
20. Veenema, O., Thatte, N., Rigopoulou, D., et al., (2026), MNRAS, 548, stag785
    Decoupling the AGN outflow and star-forming disc kinematics in the nuclear region of NGC 7582 with JWST NIRSpec and MIRI/MRS
21. Riffel, R., Souza-Oliveira, G., Colina, L., et al., (2026), A&A, 710, A231
    Jet-driven shocks and turbulence in radio-loud Active Galactic Nuclei observed with JWST MIRI/MRS
22. Pantoni, L., Baes, M., Decin, L., et al., (2026), A&A, 709, A237
    MICONIC: JWST/MIRI-MRS reveals heavily reprocessed polycyclic aromatic hydrocarbons in the circumnuclear disc of Centaurus A
23. Haidar, H., Rosario, D., García-Bernete, I., et al., (2026), MNRAS, 546, stag069
    GATOS XI. Excess dust heating in the narrow-line regions of nearby AGN revealed with JWST/MIRI
24. Pérez-González, P., Barro, G., Carniani, S., et al., (2026), ApJ, submitted
    Little Red Dots: One Photometric Tag Concealing Diverse Spectroscopic Flavors of Massive Star Formation and Black Hole Activity
25. García-Bernete, I., Pereira-Santaella, M., González-Alfonso, E., et al., (2026), Nature Astronomy, 10, 420
    Abundant hydrocarbons in a buried galactic nucleus with signs of carbonaceous grain and polycyclic aromatic hydrocarbon processing
26. Zhang, L., Packham, C., Hicks, E., et al., (2026), ApJ, 998, L32
    Evidence of Feedback Effects in Low-luminosity Active Galactic Nuclei Revealed by JWST Spectroscopy

27. Ramos Almeida, C., Asensio Ramos, A., Westerdorp Plaza, C., et al., (2025), A&A, 706, A100
Silicate emission in a type-2 quasar: JWST/MIRI constraints on torus geometry and radiative feedback
28. Veenema, O., Thatte, N., Rigopoulou, D., et al., (2025), MNRAS, 544, 3361
Shock-driven heating in the circumnuclear star-forming regions of NGC 7582: insights from JWST NIRSpec and MIRI/MRS spectroscopy
29. Hunt, L. K., Draine, B. T., Navarro, M. G., et al., (2025), ApJ, 993, 84
The Interstellar Medium in I Zw 18 Seen with JWST/MIRI. II. Warm Molecular Hydrogen and Warm Dust
30. Riffel, R. A., Colina, L., Costa-Souza, J. H., et al., (2025), A&A, 705, A59
Impact of AGN and nuclear star formation on the ISM turbulence of galaxies: Insights from JWST/MIRI spectroscopy
31. Hunt, L. K., Aloisi, A., Navarro, M. G., et al., (2025), ApJ, 992, 48
The Interstellar Medium in I Zw 18 Seen with JWST/MIRI. I. Highly Ionized Gas
32. Prince, R., Hernández Santisteban, J. V., Horne, K., et al., (2025), MNRAS, 541, 642
Echo mapping of the black hole accretion flow in NGC 7469
33. Ramos Almeida, C., García-Bernete, I., Pereira-Santaella, M., et al., (2025), A&A, 698, A194
JWST MIRI reveals the diversity of nuclear mid-infrared spectra of nearby type 2 quasars
34. Hermosa Muñoz, L., Alonso-Herrero, A., Labiano, A., et al., (2025), A&A, 693, A321
MICONIC: dual AGN, star formation, and ionised gas outflows in NGC 6240 seen with MIRI/JWST
35. Zhang, L., García-Bernete, I., Packham, C., et al., (2024), ApJL, 975, L2
Polycyclic Aromatic Hydrocarbon Emission in the Central Regions of Three Seyferts and the Implication for Underlying Feedback Mechanisms
36. Pereira-Santaella, M., González-Alfonso, E., García-Bernete, I., et al., (2024), A&A, 689, L12
H_3^+ absorption and emission in local U/LIRGs with JWST/NIRSpec: Evidence for high H_2 ionization rates
37. Haidar, H., Rosario, D. J., Alonso-Herrero, A., et al., (2024), MNRAS, 532, 4645
Dust beyond the torus: Revealing the mid-infrared heart of local Seyfert ESO 428-G14 with JWST/MIRI
38. Shen, Y., Grier, C. J., Horne, K., et al., (2024), ApJS, 272, 26
The Sloan Digital Sky Survey Reverberation Mapping Project: Key Results
39. García-Bernete, I., Pereira-Santaella, M., González-Alfonso, E., et al., (2024), A&A, 682, L5
Structures Of Dust and gAs (SODA): Constraining the innermost dust properties of II Zw96 with JWST observations of H_2O and CO
40. González-Alfonso, E., García-Bernete, I., Pereira-Santaella, M., et al., (2024), A&A, 682, A182
JWST detection of extremely excited outflowing CO and H_2O in VV 114 E SW: A possible rapidly accreting IMBH

41. García-Bernete, I., Alonso-Herrero, A., Rigopoulou, D., et al., (2024), A&A, 681, L7
The Galaxy Activity, Torus, and Outflow Survey (GATOS). III. Revealing the inner icy structure in local active galactic nuclei
42. Cackett, E. M., Gelbord, J., Barth, A. J., et al., (2023), ApJ, 958, 195
AGN STORM 2. IV. Swift X-Ray and Ultraviolet/Optical Monitoring of Mrk 817
43. García-Bernete, I., Rigopoulou, D., Alonso-Herrero, A., et al., (2022), A&A, 666, L5
A high angular resolution view of the PAH emission in Seyfert galaxies using JWST/MRS data
44. Varga, J., Hogerheijde, M., van Boekel, R., et al., (2021), A&A, 647, A56
The asymmetric inner disk of the Herbig Ae star HD 163296 in the eyes of VLTI/MATISSE: evidence for a vortex?